Techniques in Biotechnology



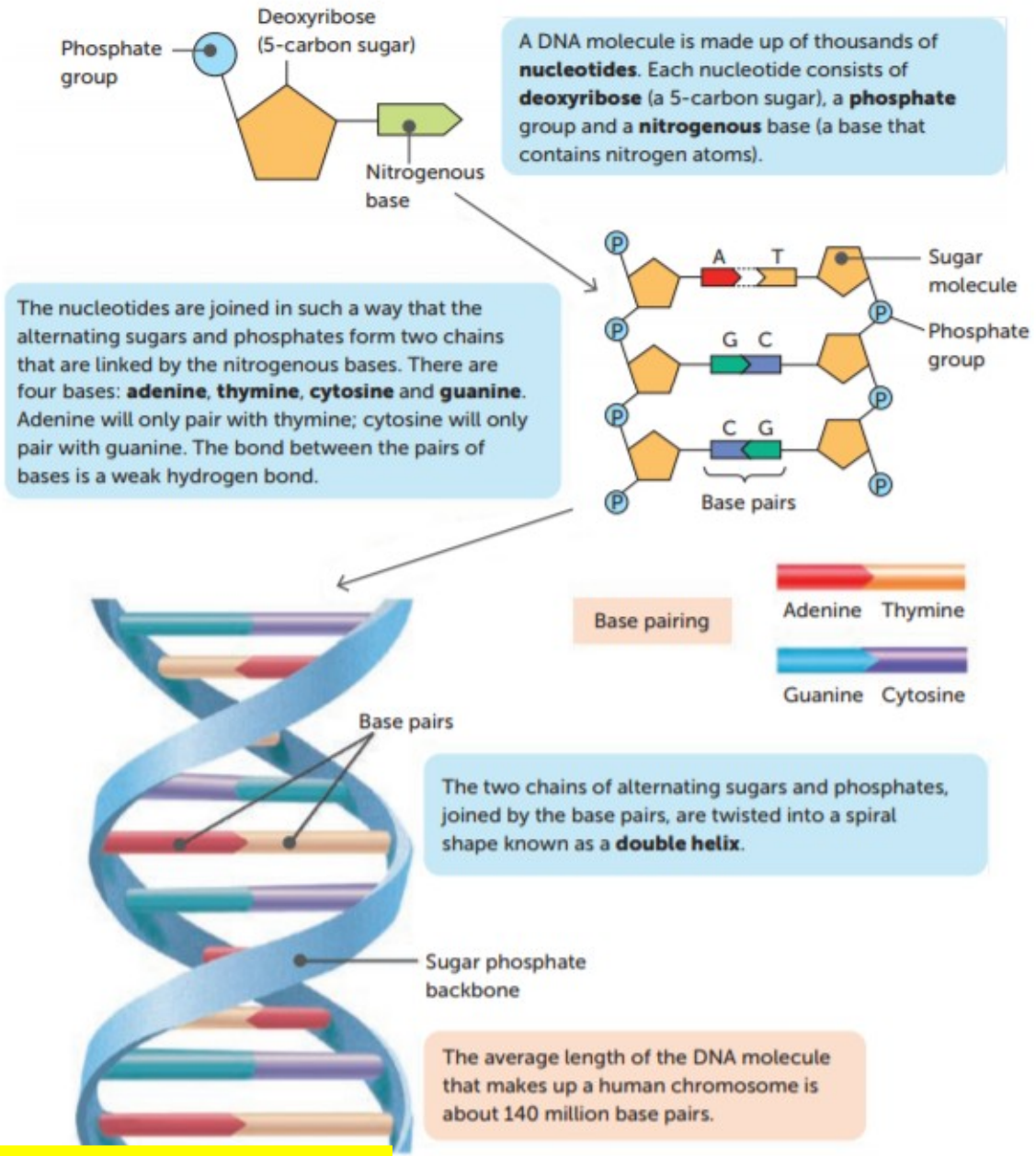
Recombinant DNA technology

Recombinant DNA Technology

- Also called “genetic engineering”
- Process of splicing DNA from one organism into another organism
- Variety of uses.
  - Synthetic hormone production
  - Transgenic organisms – eg wheat that is resistant to herbicides
  - Gene therapy for diseases (still in the research phases)

DNA Review

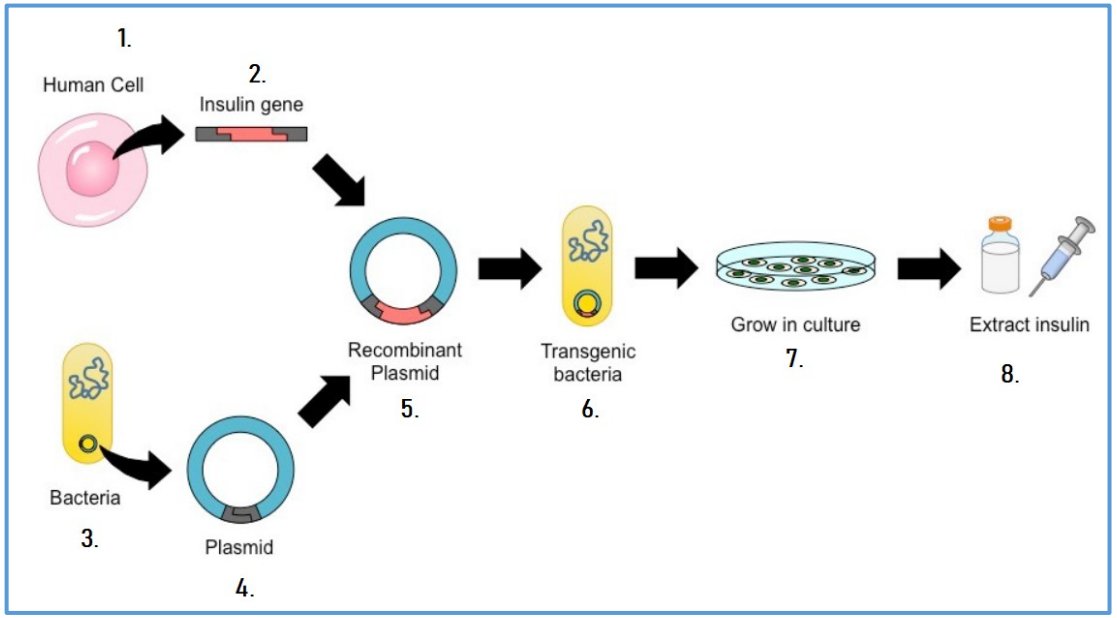
- The sequence of bases that make up the DNA code for protein production by the cell.
- A gene is a series of bases that code for the production of a particular protein.
- For example, the *insulin gene* is the sequence of bases that codes for the production of the hormone insulin, and the *growth hormone gene* is the sequence of bases that codes for the production of growth hormone.



Learning Aim: Briefly explain the role of DNA in the manufacture of proteins.

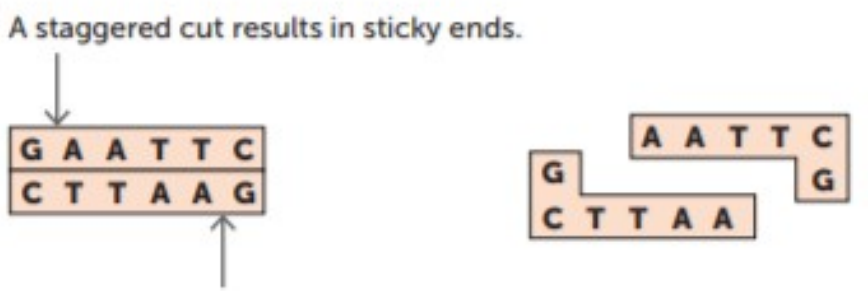
Putting a gene from one organism into another organism

- Technology is available to put a gene from one organism into a totally different organism!
- For example, scientists can put the insulin gene from a human cell into a bacterium, so that the bacterium produces the hormone insulin, which can then be extracted, refined and used to treat diabetes mellitus.
- This involves:
  - ✓ Isolating (cutting out) the insulin gene from a healthy human cell, using a *restriction enzyme*.
  - ✓ Treating a piece of bacterial DNA with the same *restriction enzyme*.
  - ✓ Splicing the human insulin gene into the bacterial DNA.
  - ✓ Sticking the spliced pieces together with the enzyme *DNA Ligase*.
  - ✓ Putting the bacterial DNA back into the bacterium.

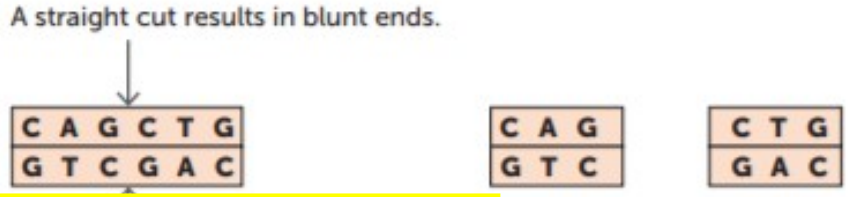


Isolating genes using Restriction Enzymes

- *Restriction enzymes* are enzymes that cut DNA when they encounter specific base sequences.
- There are different types of restriction enzymes, that target different base sequences.
- Some make *staggered cuts* with *sticky ends*.
  - They target specific base sequences (*recognition sites)* that are *palindromic* – they read the same backwards and forwards.
  - The restriction enzyme cuts along the recognition site, producing a staggered cut with sticky ends. Other base sequences can bind if they have the same staggered cut and sticky ends.
  - These can be used to isolate segments of DNA containing genes such as the insulin gene, or the growth hormone gene, and allow them to be spliced into other DNA that has the same staggered cut and sticky ends.



- Some restriction enzymes make straight cuts with blunt ends. These are not as useful in recombinant DNA technology because they are less specific in what can join with them.



Fact: Restriction enzymes are used in nature, by bacteria, to restrict viruses called *phages* from infecting them

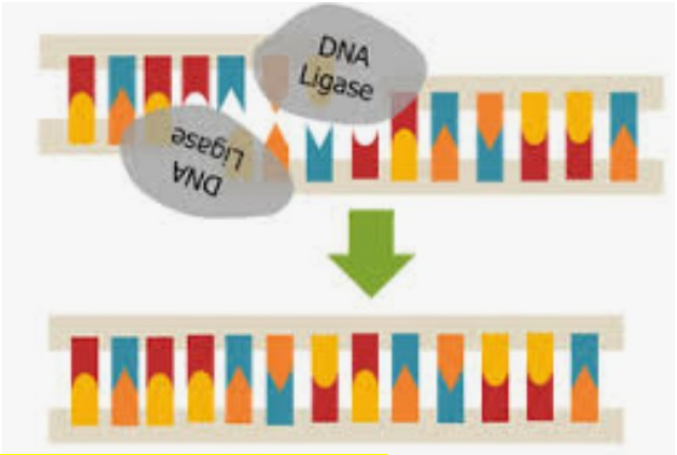
Learning Aim: Explain how Restriction Enzymes are used in Recombinant DNA Technology

Splicing two pieces of DNA together using DNA Ligase

- Two pieces of DNA that have been treated with the same *restriction enzyme* will have the same staggered cut and sticky ends - they will match even if they are from different types of organism.

- For example, if the gene for insulin production is isolated from a human cell using a restriction enzyme, it will splice into DNA from a bacterium if the two are placed together.

- Once two DNA sequences are put together with the same staggered cuts and sticky ends, they are treated with an enzyme called *DNA Ligase* which sticks them together.

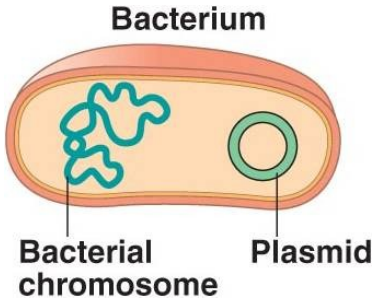


Learning Aim: Explain how DNA Ligase is used in Recombinant DNA Technology

Animation of actions of restriction enzyme and DNA ligase

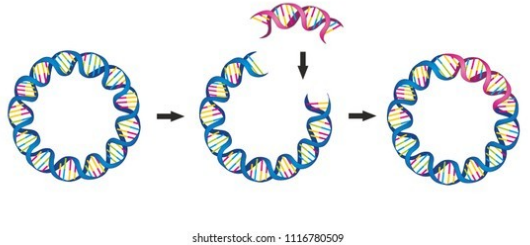
Splicing human genes into bacteria

- Bacteria have long strands of DNA, but also have smaller, circular strands of DNA in their cytoplasm.
- These small, circular strands of DNA are called *plasmids*.



- *Plasmids* are removed from bacteria and treated with the same restriction enzymes used to isolate human genes, so the human gene can be spliced into the plasmid.

- The plasmid (now containing a human gene, eg for insulin) is a *vector* (carrier) that transports the human gene.



- The plasmid is placed into a new bacterium, so that bacterium will carry the human gene. If it is carrying the human insulin gene, it will produce the hormone insulin!



Learning Aim: Explain how Bacterial Plasmids are used in Recombinant DNA Technology

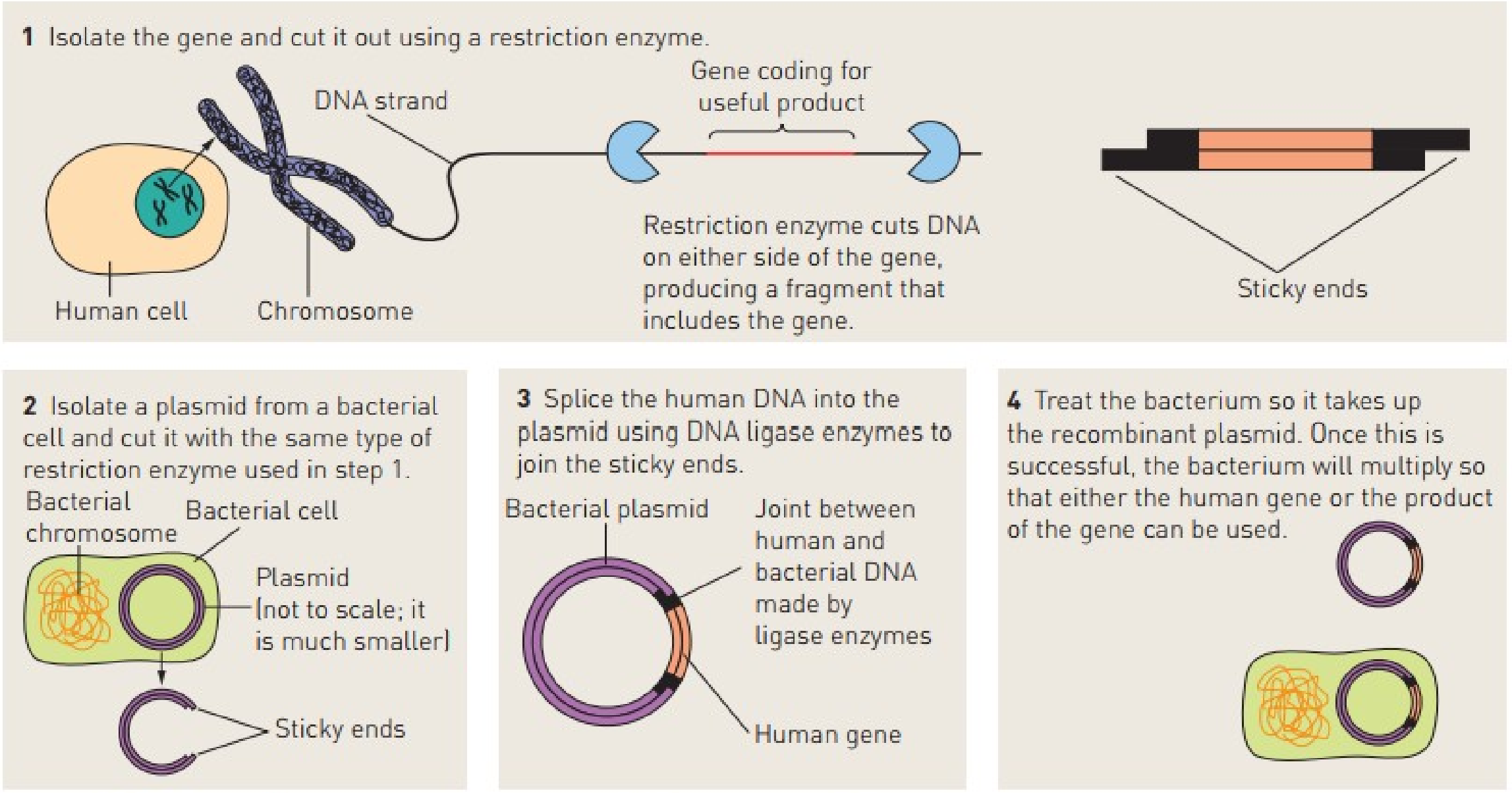
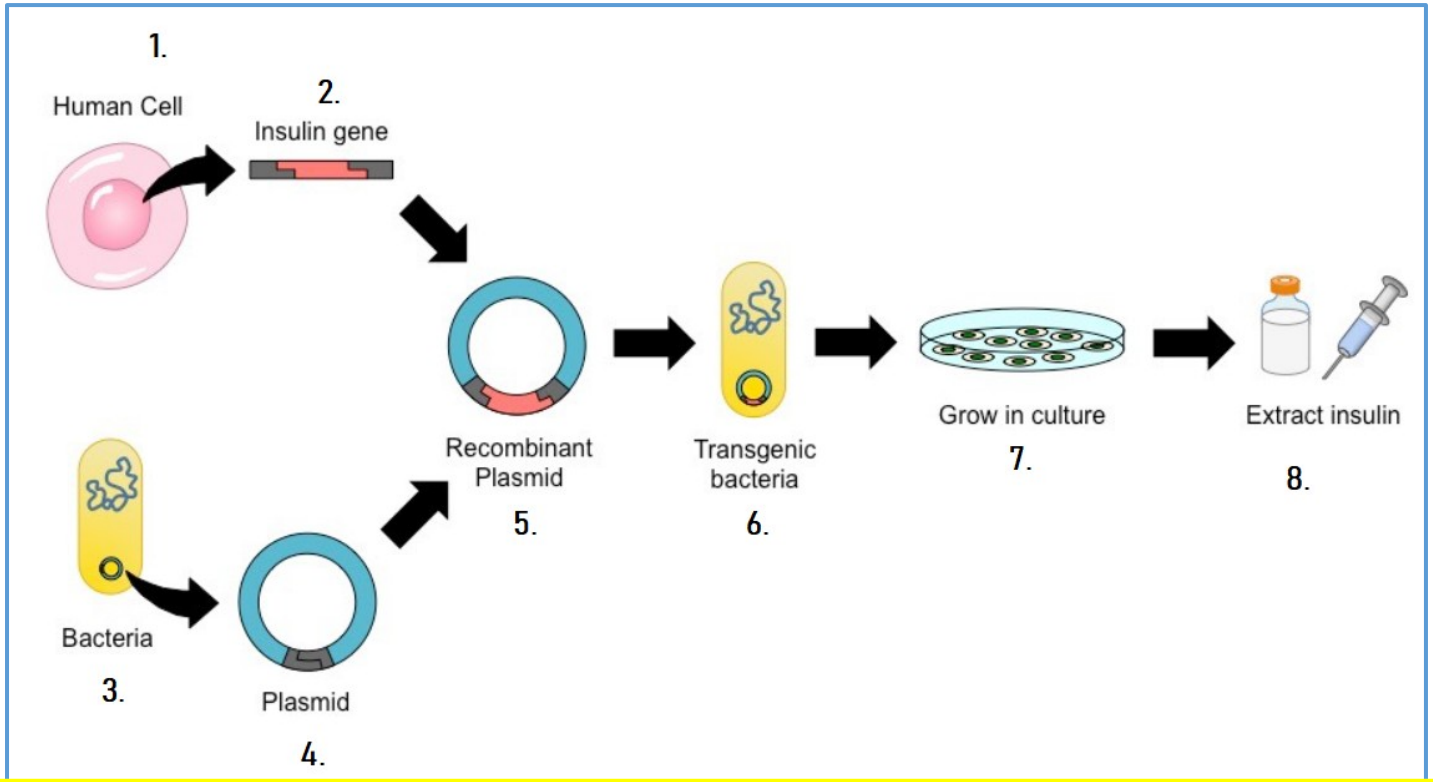


Figure 13.9 A simplified diagrammatic representation of recombinant DNA technology



Learning Aim: Describe in detail the sequence of events involved in producing synthetic insulin

Blunt ends	The ends produced by a straight cut of a sequence of nucleotide bases
Ligase	An enzyme capable of combining two small components of single-strand DNA into one single structure
Phage	Or bacteriophage; a virus that infects bacteria
Plasmid	Small circular strand of DNA distinct from the main bacterial genome; it is composed of only a few genes and is able to replicate independently within a cell
Restriction enzyme	Enzyme that cuts a strand of DNA at a specific sequence of nucleotides
Staggered cut	Produced when a restriction enzyme creates fragments of DNA with unpaired nucleotides that overhang at the break in the strands; called sticky ends
Straight cut	Produced when a restriction enzyme makes a clean break across the two strands of DNA so that the ends terminate in a base pair; called blunt ends
Sticky ends	The overhanging ends produced by a staggered cut of a sequence of nucleotide bases; can be called cohesive ends
Vector	A bacterial plasmid, viral phage or other such agent used to transfer genetic material from one cell to another

1: The base sequence coding for insulin production (the insulin gene) is isolated from a healthy human cell.

2: The base sequence on either side of the gene is treated with a *restriction enzyme* that cuts the base sequence, leaving a *staggered cut* with *sticky ends*.

3: A *plasmid* is isolated from a bacterial cell.

4: The plasmid is treated with the *same* restriction enzyme that was used on the insulin gene so that it has the *same* staggered cut and the *same* sticky ends.

5: The human insulin gene is then inserted into the bacterial plasmid, which fits because the plasmid and the gene have the same staggered cut and sticky ends. The plasmid and insulin gene are stuck together using the enzyme *DNA Ligase*.

6: The recombinant plasmid acts as a *vector* (carrier) of the human insulin gene. It is inserted into a new bacterium, creating a *transgenic* bacterium that contains the gene for human insulin. Because it contains the gene for human insulin, the bacterium can now produce human insulin hormone.

7: The transgenic bacterium is grown in culture, where it reproduces until there is a large number of bacteria, all producing the human insulin hormone.

8: The bacterial culture is processed to extract the hormone insulin that the bacteria have produced and it is refined for human use, to treat Type I diabetes.

Examples of use

- Insulin
  - Pre-1982, diabetics treated with insulin from pig/cattle pancreas
  - 1982: insulin produced from genetically engineered bacteria approved – “Humulin”
  - Human gene spliced into bacterial cells, so bacteria would then produce insulin
  - Insulin now produced in huge vats of bacteria.

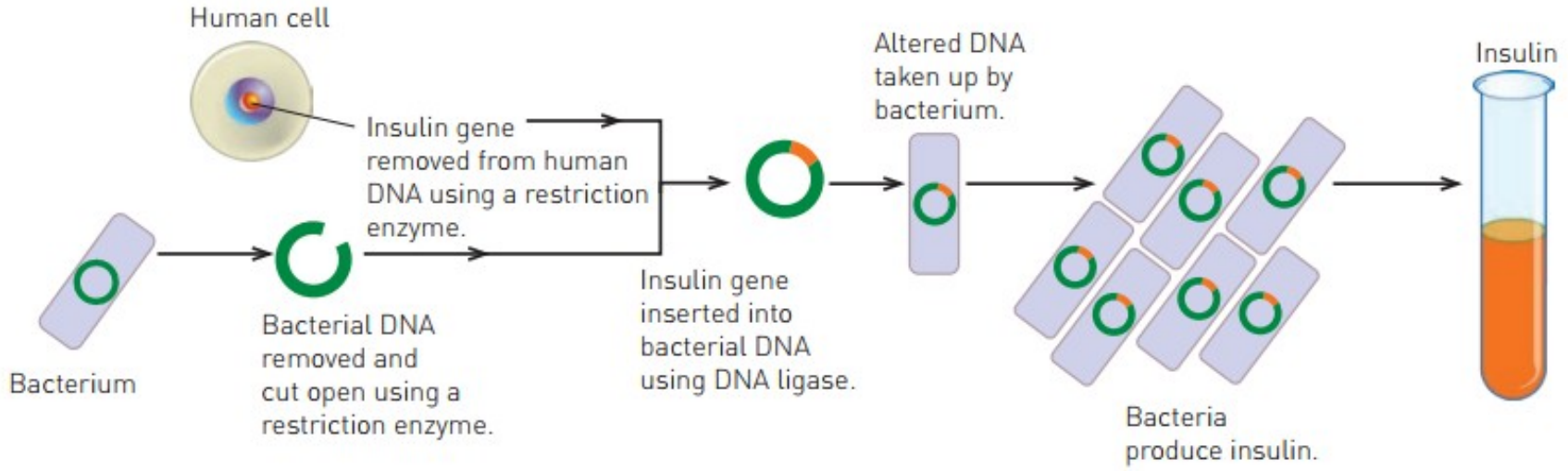


Figure 13.10 A diagrammatic representation of the production of human insulin using recombinant DNA technology

Examples of use continued

- Human Growth Hormone
  - Now produced by genetically engineered *E. coli* bacteria
  - Used to treat lack of growth hormone in pituitary disorders
  - Also used in dairy cattle to increase milk production

Note: the process for making synthetic Human Growth Hormone using Recombinant DNA technology is exactly the same as for making synthetic insulin, except the gene for Growth Hormone is isolated and worked with, not the gene for insulin.

Learning Aim: Describe in detail the sequence of events involved in producing synthetic insulin